

6	(a) (coulomb is) ampere second	B1	[1]
	(b) (total) charge or $Q = nAle$	M1	
	$I = Q/t$ <u>and</u> $l/t = v$	M1	
	$I = nAle/t = nAve$ therefore $v = I/nAe$	A1	[3]
	(c) (i) ratio = $(I/nA_Ye)/(I/nA_Ze)$	C1	
	$= A_Z/A_Y$ or $4A/A$ or $\pi d^2/(\pi d^2/4)$	C1	
	$= 4$	A1	[3]
	(ii) $R = \rho l/A$ or $R = 4\rho l/\pi d^2$	B1	
	$R_Y = \rho l/A$ <u>and</u> $R_Z = \rho(2l)/4A$ so $R_Y/R_Z = 2$ or $R_Y = 4\rho l/\pi d^2$ <u>and</u> $R_Z = 4\rho(2l)/\pi 4d^2$ or $2\rho l/\pi d^2$ so $R_Y/R_Z = 2$	A1	[2]
	(iii) $V = 12R_Y/(R_Y + R_Z)$ or $I = 12/(R_Y + R_Z)$ <u>and</u> $V = IR_Y$	C1	
	$V = 12 \times 2/3$ $= 8(.0)V$	A1	[2]
	(iv) ratio = I^2R_Y/I^2R_Z or $(V_Y^2/R_Y)/(V_Z^2/R_Z)$ or $(V_YI)/(V_ZI)$		
	$= 2$	A1	[1]

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